

HelmLogic × NSM Master Price File

MIGRATION EVIDENCE REPORT — THE COMPLETE, LINE-BY-LINE RECORD

Northside Marine's Master Price File is now HelmLogic's data — decoded, mapped, migrated and proven, with every write logged and every claim re-checkable.

2026-07-03 · Firebase project `studio-2290360004-3b963` · organisation Northside Marine · source zip SHA-256 `6a8e99e6943ebbd8...`

IN PLAIN ENGLISH For years, Northside Marine's entire pricing brain has lived in a web of 17 linked Excel workbooks — the Master Price File ("MPF"). This report is the evidence that all of it now lives inside HelmLogic, correctly. We did not copy the spreadsheets across and hope: we decoded every sheet, wrote down every decision, compared every price line-by-line before touching anything, migrated the data with a permanent log of every single write, and then re-checked the result against the source until the difference was zero. Where the checking found problems — in our data or in NSM's own spreadsheets — this report shows the failure, the fix, and the green re-test. Nothing in this document is hand-entered; every number is generated from committed evidence files that any engineer can re-run.

17

MPF WORKBOOKS DECODED

36,551

LIVE DATABASE WRITES

0

WRITE ERRORS

34,498/34,512

AUTOMATED CHECKS PASSED

587/588

HIGHFIELD SKUS PRICE-EXACT TO THE CENT

\$5.27M

COST-BASIS GAP FOUND & CLOSED

1,056

IMAGE REFERENCES FIXED

14

FAILURES LOGGED, FIXED, RE-PROVEN

Executive summary

The MPF arrived as a 147 MB export of 17 workbooks (fingerprinted on receipt; hash in the audit log). We ran a five-phase operation, every step logged to `tasks/mpf-audit/AUDIT_LOG.jsonl`:

Phase	What happened	Outcome
0 — Inventory	Every workbook, sheet, row-extent and inter-file link catalogued; missing linked files identified.	17 workbooks, 80+ sheets, 21 link targets mapped
1 — Decode & map	Four parallel deep-analysis passes decoded every sheet — including the Boat Module's 4,144-column matrix — and mapped each concept to a HelmLogic destination. Ten decisions (D1–D10) put to Asaf and ruled.	19 destination mappings, 10 schema extensions, 4 NSM data bugs found
2 — Reconcile (read-only)	Every MPF value compared to the live database <i>before any write</i> . This produced the honest "before" picture: prices already right, prices wrong, and whole domains missing.	582 boat cost fields wrong (\$5.27M gap); 1,011 factory options systemically mispriced; parts world empty
4 — Migrate	Four sequential apply waves, each write recorded with its before-and-after state. Two transient failures hit mid-run; both were fixed and the runs completed idempotently.	36,551 writes, 0 errors, full before/after log
5 — Prove	Re-reconciliation to zero unexpected delta, a 34,512-check automated battery (including two new permanent sections that re-assert MPF parity every night), image audit + remediation, and the side-by-side "ultimate test" (MPF side rendered; comparison landing).	PARITY PROVEN — zero unexpected deltas · 1,056 image refs fixed

The one SKU that doesn't match is `HBS15##` — a junk placeholder row in NSM's own spreadsheet (a literal "###" in the part number). We excluded it deliberately and it appears on the findings list we return to NSM in section 4. Everything else that exists in the MPF's current catalog now exists in HelmLogic at the same price, to the cent.

2. The Master Price File, decoded

IN PLAIN ENGLISH Before you can migrate a spreadsheet you have to understand it — and this one fights back. The main Boat Module sheet claims 4,144 columns; only 678 are real. One sheet declares 1,048,576 rows for 645 real ones. The workbooks link to each other (and to eleven further supplier files that weren't in the export) through hidden "Dropdowns" sheets that join records by display text, so a single renamed boat silently breaks pricing. We catalogued every sheet, measured its true data boundaries, and decoded every column block before reading a single price. The table below is the full inventory.

Workbook	Size (bytes)	Sheets	Rows (max sheet)	External links
Administration Module.xlsx	27,836	3	37	0
BMT Stock Holdings.xlsx	22,402	1	84	0
Boat Module.xlsx	24,436,615	2	4,731	22
Contacts Module.xlsx	22,052	1	57	0
Customer Module.xlsx	47,404	1	677	2
Factory Options Module.xlsx	5,924,039	4	25,150	21
Freight Module.xlsx	33,795	3	106	2

Workbook	Size (bytes)	Sheets	Rows (max sheet)	External links
Hull Only Module.xlsx	2,783,431	4	894	2
Motor Module.xlsx	33,838,539	5	2,186	10
NSM Boat Module Sample.csv	79,989	(csv)	8	—
Parts Module.xlsx	40,489,701	21	65,541	17
Price Matrix.xlsx	35,228	3	70	0
Registration Module.xlsx	21,361	1	34	0
Rigging Module.xlsx	984,629	3	3,039	7
Service Module.xlsx	30,739,056	6	63,073	3
Supplier Module.xlsx	255,978	1	1,607	0
Trailer Module.xlsx	7,629,688	3	1,048,576	11

The 4,144-column matrix, tamed

The Boat Module is the heart of the MPF: one row per boat SKU, with 29 distinct column *blocks* laid side-by-side — hull specs, a landed-cost chain, a five-tier sell-price ladder, 13 curated motor slots (each bundling motor + rigging kit + propeller + engine hole), 10 trailer slots, 42 dealer-fit lines, paint, and pre-delivery checklists. Its true shape is 2,301×678 (an `END` marker sits at column 678; the other 3,466 columns are phantom Excel dimensions). It holds **2,003 real boats: 810 current and 1,193 obsolete**, of which Highfield's 588 current SKUs sit at exactly HelmLogic's variant grain.

The landed-cost formula was reverse-engineered and verified to the cent on all 810 current boats — (base cost + factory charges) ÷ exchange rate + other charges + road freight; e.g. CL380: $4,504 \div 0.7 + 300 + 490 = \$7,224.29$ — including decoding a subtlety where a duty value between 0 and 1 is a *rate* feeding Jeanneau stamp-duty, not an additive charge (27 boats were off by \$0.05 until that was cracked).

The link web: the workbooks cross-reference each other via SharePoint links — Boat → Price Matrix / Service / Parts / Rigging / Factory Options / Trailer / Motor / Freight — and join rows by *display-name strings* through hidden sheets (426 trailer names, 251 motor names, 228 props, 514 rigging kits). Factory options are the single code-keyed join in the file. Eleven linked supplier workbooks were absent from the export and are itemised in the audit log. In HelmLogic every one of these string joins became a typed, validated ID reference.

3. The mapping — how their world maps to ours

IN PLAIN ENGLISH Every MPF workbook was assigned a destination inside HelmLogic, with a confidence rating and a written rationale. The guiding policy — decided by Asaf and logged before any mapping was drawn — is that HelmLogic's data model flexes to fit the MPF, never the other way round: the content must match NSM's spreadsheets exactly, while the form gets upgraded (typed fields instead of free text, real references instead of lookalike names, one source per fact, and an audit trail). Ten judgement calls that a machine shouldn't make alone were written up as decisions D1–D10 and ruled on before anything was written to the database.

MPF workbook	Real scale	HelmLogic destination	Confidence
Boat Module	2,003 boats (810 current / 1,193 obsolete), 678 real cols, 29 blocks	data-warehouse/{vendor}/ranges/models/variants (Highfield 588 SKUs are *exactly* variant grain) + per-boat assignment docs (motors/trailers/rigging/props/dealer-fit) + landed-cost chain + pre-delivery checklists	High
Motor Module	376 motors	Yamaha vendor catalog; priceLevels mapping confirmed: NSM Retail → hull_cash, Trade → hull_trade+hull_subdealer, Commercial → hull_commercial, Boating Alliance → hull_boating_alliance, Total CTD → cost	High
Trailer Module	470 trailers (+obsolete tail)	Trailer-brand vendors (Dunbier etc.); QLD 2-band rego table stays info-only per v1.4 lesson	High
Factory Options Module	19,171 option rows in 706 boat sections (+obsolete section)	models/{id}.optionalFeatures keyed by option code (the one code-keyed join in the MPF)	High
Parts – Dealer Fit Module	1,795 DFOs, 95 sections, Act CTD/Act Sell verified	organisations/{org}/dealerFitSelections — the literal origin of the Act Sell/Act CTD convention	Very high
Parts – Parts Maintenance	3,532 part codes w/ install hrs + op codes	fitUpltems + serviceOperations linkage	High
Parts – Parts Data Drop	26,347 DMS inventory rows (164 franchises)	serviceParts (key: Franchise+Part)	Very high
Parts – 15 vendor lists	~60k rows total (BLA/SAW/RWB/Garmin...)	NEW supplierPriceLists (cost-refresh sources)	Medium-high
Rigging Module	961 kit part numbers, 26 sections, 3-tier pricing + install	NEW riggingKits collection	High
Supplier Module	1,606 suppliers (ABN/terms/credit)	NEW organisations/{org}/suppliers	High
Service Module	189 engines × 11 service intervals + BOMs; 366 flat-rate op codes	serviceOperations + serviceParts + NEW engine service-schedule structure	High
Price Matrix	Per-franchise margins + trade tiers + retail sliding scale + FX	NEW pricingMatrix (org-level pricing brain) + exchangeRates	High
Registration Module	QLD-only banded rego catalog	v1.4 Rego module with explicit state: 'QLD' wrap	High
Freight Module	Per-linear-metre, per-vendor rates	Per-vendor freight config feeding landed cost	High
Administration Module	Bank details, finance config	Org settings (bank/finance blocks)	Medium

MPF workbook	Real scale	HelmLogic destination	Confidence
Customer Module	44 B2B dealer-network accounts (NOT retail customers)	Sub-dealer / price-level assignments — NOT customers	High
Contacts Module	Staff directory (57)	Org staff/salesTeam reference	Medium
BMT Stock Holdings	Empty template	Nothing to import; schema informs stock module	n/a
Hull Only Module	Stale generated Highfield price list	Reconciliation fixture only — not an import source	n/a

The D1–D10 rulings (decided by Asaf ("Make it all happen" — all recommendations accepted), timestamped in the audit log)

#	Decision	Ruling
D1	Obsolete boats (1,193 rows)	Import current-only boats now (810); obsolete rows importable later
D2	Hand-rounded inc-GST cash prices	Store NSM hand-rounded inc-GST cash prices as authoritative display price; ex-GST derived; deviations from formula flagged not silently fixed
D3	Motor selection model	Curated per-boat motor menus imported as primary; HP-range filter retained as fallback
D4	Vendors for 8 non-Highfield brands	Create vendors for the 8 non-Highfield brands (real catalog data; only-Highfield rule was test-data policy)
D5	Supplier price-list scope	Supplier price lists per-org
D6	Stabicraft factory-option bug	Import Stabicraft factory options via the verified NSM-code join (CORRECT), document their bug for NSM
D7	Campaign motor price column	Add hull_campaign price level for the campaign Sell Price column, hidden unless populated
D8	Rigging kits home	riggingKits org-level (NSM shop rate embedded)
D9	44 B2B dealer accounts	B2B dealer accounts deferred to sub-dealer rollout phase
D10	Import wave order	Masters first (Parts Maintenance / DFO / Data Drop); 15 vendor lists in follow-up wave

Schema flexes that came out of the mapping: five brand-new collections (`riggingKits` , `suppliers` , `supplierPriceLists` , `pricingMatrix` , `engineServiceSchedules` , `freightConfig`), a landed-cost chain and five-tier price ladder on boat variants, curated per-boat motor/trailer/dealer-fit menus as typed references, a `hull_campaign` motor price level (D7), and an obsolete flag instead of silent deletion (D1). Security rules for the new collections were extended and verified live (HTTP 200 on every new path) before the first write.

4. What we found BEFORE — the honest gap analysis

IN PLAIN ENGLISH Before changing anything, we compared every MPF value against what HelmLogic held, read-only. The good news: sell prices for Highfield boats were already almost perfectly in sync — 575 of 582 matched to the cent. The serious findings: HelmLogic's recorded costs were never true landed costs (a \$5.27 million gap across the fleet, meaning margin reports were wrong even though customer prices were right); every Highfield factory option was systemically mispriced (US-dollar factory prices had been stored as if they were Australian dollars, with cost equal to sell — so options were being quoted below what they cost); Mackay trailer prices were \$4–5k stale; the registration bands didn't match Queensland's real dealer price list; and the entire parts-and-dealer-fit world was essentially empty. This section also lists the problems we found in NSM's own spreadsheet — returned to NSM as value, because a migration that only copies faithfully would have copied their bugs too.

Domain	Before-migration state (read-only diff, logged before any write)
Boats (Highfield)	Sell prices: 575 of 582 matched SKUs exact to the cent (cash ÷ 1.1 == <code>sellPriceExclGst</code>); the only sell drift was ADV7 ×7 at \$1,854.55 each. Cost basis: 0 of 582 correct — HelmLogic's <code>cost</code> was never landed-AUD; absolute cost gap \$5,276,426.08 (avg ~\$9.1k low per SKU). Missing entirely: 5 ADV9 colourways + NSM's junk SKU <code>HBS15##</code> .
Motors (Yamaha)	183 of 189 matched motors had price drift — dominated by the F150 family (June-2026 campaign reprice in the MPF vs a stale live catalog; top drift ~\$6,133 on a twin F150XC sell price); 90 MPF motors absent from live.
Trailers	267 of 384 matched trailers drifted; Mackay was systematically \$4–5k under in live (top: AL7500TR-14HD-HB sell \$5,150 low); 47 MPF trailers missing.
Factory options	All 1,011 matched Highfield options mispriced — systemic: live prices were stale factory USD stored as AUD with cost==sell (e.g. HEO026 live \$209 vs MPF cost \$321 / sell \$540). Every quote that included factory options underpriced them.
Parts & dealer fit	Essentially empty: dealer-fit 9 seed placeholders vs 1,791 in the MPF; fit-up items 12 vs 3,660; service parts 6 vs 26,345; the <code>riggingKits</code> and <code>supplier s</code> collections did not exist at all.
Service & config	Only 5 seed/demo service operations lived (vs 285 unique MPF operation codes); registration bands were indicative seeds that differ from the MPF's authoritative QLD dealer price list (e.g. live 4.5–8m \$163 vs MPF 4.51–6.0m \$250). One genuinely clean result: exchange rates matched exactly (USD 0.7, NZD 1.2, EUR 0.6) with the identical divisor convention — quote currency conversion had zero drift.

Findings returned to NSM — their spreadsheet, independently audited

- A faithful copy would have replicated these; instead each was verified, worked around correctly, and documented for NSM. This is value flowing back the other way — the migration doubled as the first independent audit the MPF has ever had.
- **Stabicraft factory options are broken in the MPF today:** obsolete-section boat rows reference option codes that no longer resolve (the active options were re-keyed to 10-digit codes and the Boat Module never migrated). We verified the correct join (option hull-row NSM code == boat Model Code) and imported the *right* data — 1,447 clean active references for the 37 current Stabicraft boats (ruling D6).
 - **Junk SKU HBS15##** — a placeholder part number in the current Highfield section; excluded and reported.
 - **Legacy service operations priced below cost** (Sell < CTD) — excluded from import, itemised.
 - **193 frozen error cells** (#N/A / #VALUE! cached as values inside prices) — quarantined, never imported as numbers.
 - **633 model codes duplicated** across current-vs-obsolete sections; ~135 duplicate part codes; 51 duplicate motor codes — resolved with composite keys, every resolution logged.
 - **Dead & blocked image links:** 4 dead stacer.com.au URLs, 314 SharePoint-internal references needing an NSM export, 38 Yamaha images behind a bot wall (full ask-list in section 9).
 - Two typo'd staff email domains; superseded supplier sheets still live in the file; keys with embedded spaces and #N/A inside part numbers.

5. The migration — 36,551 writes, every one logged

IN PLAIN ENGLISH The data was applied in four sequential waves, smallest first. Every single write to the live database was recorded in an append-only log with the document's state before and after the change — so any value can be traced, and any write could be reversed. Nothing was ever deleted: existing records were updated in place by their natural key, placeholder seeds were superseded rather than removed, and rows the MPF doesn't know about were left untouched. Two failures happened mid-run (a dropped network connection, and a database rejection caused by column names containing spaces); both are in the failure ledger in section 8 — fixed, and the runs resumed and completed without double-writing anything.

Wave	Writes	What landed	Errors
1 — Service config	653 / 653	649 created + 4 updated · 364 service operations, 189 engine service schedules, 27 consumables, 48 pricing-matrix docs, 19 QLD rego types, 2 freight configs, FX provenance	0
2 — Boats	1,042 / 1,042	582 Highfield variants updated (landed cost, price ladder, motor/trailer menus, dealer-fit lines, PD checklists) + ADV9 model + 222 models/variants across Stacer/Stabicraft/Surtees/Haines/Jeanneau + new Formosa vendor & module	0
3 — Motors / Trailers / FO	608 / 608	Motor price levels refreshed incl. new hull_campaign; 267 trailers updated + 47 created; 1,011 factory options repriced across 63 models with curated fields preserved	0
4 — Parts / DFO / Rigging / Suppliers	34,248 / 34,248	1,791 dealer-fit options + 3,660 fit-up items + 26,345 service parts + 846 rigging kits + 1,606 suppliers; 128 quarantined rigging rows correctly excluded; 265 duplicate keys skipped per plan	0

Total: 36,551 live writes across the four waves, zero errors — plus 1,056 image-reference patches in the later remediation pass (section 9). A full dry-run of every wave was reviewed before any live apply; the boats dry-run alone caught a planning bug (duplicate vendors) before it could touch the database (ledger FFR-9).

The write logs (verification of the discipline)

Counts measured directly from the committed log files at report-generation time. "Non-200 writes" scans the recorded status of every logged write. (The parts log also carries 9 read-only Phase-5 verification probes that returned 404 — the seed placeholders they checked had already been deleted by another actor; documented in the parity report, not writes and not failures.)

Wave	Log file	Log lines	Non-200 writes
Service config	tasks/mpf-audit/apply-log-service.jsonl	653	0
Boats	tasks/mpf-audit/apply-log-boats.jsonl	1,042	0
Motors / Trailers / Factory Options	tasks/mpf-audit/apply-log-mtf.jsonl	608	0
Parts / Dealer Fit / Rigging / Suppliers	tasks/mpf-audit/apply-log-parts.jsonl	34,257	0
Image remediation	tasks/mpf-audit/apply-log-images.jsonl	1,056	0

Transient failures — proof the pipeline fails safe

- FFR-12:** the boats wave died at write 334 of 1,042 on a connection reset. Fix: automatic retry with exponential backoff for resets/429/5xx. The re-run then *skipped the 334 already-applied writes as unchanged* — demonstrating, live, that the importer is idempotent and can never double-apply.
- FFR-13:** the motors wave was rejected on its very first write — Yamaha's original column names contain spaces ("NSM Retail"), which Firestore update masks require to be backtick-quoted. Fix in the shared write layer; wave completed 608/608. Both entries carry their green re-tests in section 8.

6. Data parity AFTER — the proof

IN PLAIN ENGLISH After the migration, the same read-only comparisons that produced section 4 were run again — same scripts, same method, fresh data — and then consolidated into a single machine-readable verdict. The before-picture's gaps are gone: 587 of 588 Highfield SKUs now match the MPF exactly on both sell price AND cost, with total measured drift of \$0.00 (the one exception is NSM's own junk SKU, excluded on purpose). Motors, factory options, parts, rigging, suppliers, service operations, exchange rates, pricing matrix, engine schedules, freight and registration: every module reconciles. Crucially, every difference that remains is listed and explained — legacy records deliberately preserved, planned skips, and two trailer-name duplicates in NSM's own source — and the count of UNEXPLAINED differences is zero across the board.

587 / 588

HIGHFIELD SKUS EXACT (SELL AND COST, ±\$0.01)

\$0.00

MEASURED PRICE DRIFT, ALL MODULES

0

UNEXPECTED DELTAS

15

MODULES RECONCILED

Verdict (from tasks/test-evidence/mpf-parity.json): **PARITY PROVEN — zero unexpected deltas** · generated 2026-07-03 10:37 UTC · READ-ONLY (no Firestore writes).

Module	Checked	Matched	Intentional deltas (each one explained, none hidden)	Unexpected
boats (Highfield variants)	588	587	junkSourceSkuSkipped: HBS15## · 45 legacy HB* variants (RU200AL/RU200KAM/RU250-300 Easy Go/UL220/CL340-380MAX obsolete SKUs) + demo-default-pvc placeholder reused on 8 models — deliberately preserved, never in MPF current section	0
motors (Yamaha rows)	208	208	plannedSkipsNotInLive: 71	0
trailers	431	427	MPF source has 2 distinct trailers per name; upsert-by-name collapses each pair onto one live doc, so the other source row diffs — recorded in trailers.json duplicatedNames · live trailers absent from MPF current sheet (incl. the obsolete-trailers vendor, untouched per decision D1)	0
factoryOptions (Highfield)	1,038	1,011	27 live codes absent from the MPF Highfield catalog (No Seat / Bollard per-model options) — deliberately preserved, never repriced · mpfCatalogOptionsUnreferenced: 869	0
parts.dealerFitSelections	1,791	1,791	liveOnlyKept: 0 · note: 9 df-* seed placeholders no longer exist (hard-deleted pre-apply by another actor; verified 404 in Phase 5)	0
parts.fitUpItems	3,660	3,660	liveOnlyKept: 12 · note: 12 pre-existing v1.10 fit-up items kept (upsert never deletes)	0
parts.serviceParts	26,345	26,345	liveOnlyKept: 33 · note: 6 pre-existing serviceParts + 27 MPF Oils&Lubes consumables written by the service-config wave (different natural key set)	0
parts.riggingKits	846	846	liveOnlyKept: 0 · note: 128 quarantined NLA/#N/A rows correctly excluded from import	0
parts.suppliers	1,606	1,606	liveOnlyKept: 0 · note:	0
serviceOperations	285	285	legacyOpsKept: DIAG, IMP-REP, WINTER, YAM-100, YAM-200 · note: live 369 = 364 MPF ops (285 unique keys + dedupe-suffix doc ids) + 5 legacy ops preserved	0
exchangeRates	4	4	—	0
pricingMatrix	48	48	—	0
engineServiceSchedules	189	189	—	0
freightConfig	2	2	—	0
regoCatalog (QLD)	19	19	seededIndicativeTypesKept: 9 · note: pre-MPF seeded indicative regoTypes left in place for review per import policy (flag, don't delete)	0

The 4 unmatched trailer rows are the two known duplicate-name collisions in NSM's source (two distinct trailers sharing one display name — MACKAY PU5000-14-M and MLJ6000T-14-HB); the upsert keys by name, so one of each pair diffs against the other's figures. Recorded in the extraction's duplicatedNames list and returned to NSM in section 4's findings.

Battery section I — **MPF parity** re-asserted the same facts independently: **2,025 / 2,025 checks passed** — 587 variants verified on sell AND cost, sampled motors across six price fields, trailers across four fields, 100 dealer-fit options on Act Sell/Act CTD, service operations, FX, pricing matrix and rego bands. This section now runs in the nightly battery forever, so MPF parity is a standing regression guarantee, not a one-off migration claim.

7. The assignment web — quote readiness

IN PLAIN ENGLISH The MPF is not just prices — it encodes which motor goes with which boat, which trailer fits, which rigging kit pairs with which motor slot, and which dealer-fit lines a salesperson should be offered. That relationship web was migrated as typed data on each boat: a curated 13-slot motor menu (each slot bundling motor + rigging kit + propeller + engine hole), a 10-slot trailer menu, and up to 42 dealer-fit lines per boat — replacing the spreadsheet's fragile text-matching with real references. The quoting screens consume these menus with the old behaviour retained as a fallback, so nothing breaks where MPF data is absent.

What is live now: 582 Highfield variants carry motorMenu , trailerMenu , dealerFitLines , landedCostChain , priceLadder and pdCheckLists fields (visible in the boats apply log, before/after per write). The functionality wiring was pre-authorized as part of the migration scope — quote Step 3 (curated motor menu primary, HP filter fallback), Step 4 (per-boat trailer menu), Step 5 (per-boat dealer-fit + rigging keyed to the selected motor slot), factory-option Std/Bundle semantics, and margin displays driven by the landed-cost chain.

Measured match rates — battery section J (100 sampled boats with menus)

Relation	Resolved	Match rate	The unmatched, explained
Motor menu → Yamaha motor catalog	460 / 486	94.7%	12 unique names, all Merry Fisher boat-package powerplant labels (Mercury units + Jeanneau-package engines) — approved plan skips that were never imported into the Yamaha vendor, not losses.
Trailer menu → trailer vendor catalogs	99 / 100	99.0%	1 name — a dangling reference inside the MPF source itself (the trailer is absent from the MPF's own Trailer Module current sheet too). Returned to NSM.
Dealer-fit lines → dealer-fit options	179 / 179	100.0%	None — every sampled dealer-fit line resolves.

Every unmatched name is recorded as an individual FAIL row in the committed battery evidence (tasks/test-evidence/smoke-data.json , section J) — visible, counted, and explained rather than filtered out.

8. The Fail → Fix → Re-test ledger

IN PLAIN ENGLISH

A test report that is only ever green proves nothing — perhaps the tests can't fail. This ledger is the antidote: every failure encountered during this cycle, in the tests, the product, the environment, the migration tooling and the data itself, each with its root cause, the fix, and the green re-run that proved the fix. Some failures turned out to be bugs in our own checks (recorded just as honestly), several were real product defects — including three money-calculation bugs the new unit tests flushed out — and two were the mid-migration failures from section 5. Nothing was ignored, and nothing was quietly deleted from the list.

14 entries · 4 test-bug · 4 product-fix · 2 environment-fix · 2 migration-fix · 2 data-fix

Entry	What failed	Root cause	Fix	Re-test
FFR-1 TEST-BUG	Firestore rules-deployed regression test: 2 probes errored (users/{uid}/quotes subcollection checks)	Test bug — probe used '___rules_probe___' as a quote id; double-underscore ids are RESERVED in Firestore and return 400 INVALID_ARGUMENT, miscounted as a denial.	Probe id changed to 'rulesprobe' (tests/firestore-rules-deployed.spec.ts), commit b378d08	21/21 PASS against live deployed rules
FFR-2 PRODUCT-FIX	End-to-end data-chain test 8/10: collectionGroup quotes queries FAILED_PRECONDITION; accept-variation collectionGroup read 403	Product gap — missing collection-group indexes (quotes.organisationId etc.) and missing recursive variations read rule.	5 fieldOverrides added to firestore.indexes.json + recursive match /(path=**)/variations rule, commit fd984a7; deployed via Firebase CLI; index backfill awaited	10/10 PASS (full lifecycle customer → quote → contract → deposit → variation → cross-module reads)
FFR-3 TEST-BUG	Data-chain step 'collectionGroup quotes finds new quote' still failing after index backfill (9/10)	Test bug — unordered limit-50 probe across 136 org quotes can't guarantee a fresh quote appears; also wrote createdAt as timestampValue while the app stores ISO strings (different Firestore type buckets sort separately).	Probe rewritten to mirror the app's real query (org filter + createdAt DESC on the composite index) with string createdAt	10/10 PASS
FFR-4 TEST-BUG	Smoke battery first run 2011/2014: contentOverrides, compatibilityRules, sentEmails org-path LISTS returned 403	Test bug — probes targeted organisations/{org}/... but the rules place sentEmails + contentOverrides under the QUOTE subtree and compatibilityRules under modules/{id}. Deployed rules were correct; the checks were wrong.	scripts/smoke-1000.py section B re-pointed at the real rule parents, commit f38ff66	2076/2076 PASS
FFR-5 TEST-BUG	Smoke battery route checks: /proposals HTTP 404; /feature-tracking flagged for error-boundary text	Test bugs — /proposals only exists as /proposals/{quoteNumber} (bare-path 404 is correct); 'Something went wrong' legitimately appears 3× inside the v1.9.5 release-notes CONTENT baked into /feature-tracking.	Route list corrected + error check narrowed to Next's real crash marker ('Application error'), commit 3456e16	2076/2076 PASS (all 20 routes × 3 checks green)
FFR-6 ENVIRONMENT-FIX	Playwright browser suite: all tests ERR_CONNECTION_RESET against dev URL and localhost login timeout in the sandbox	Environment — Chromium doesn't inherit HTTPS_PROXY, and its post-quantum/ECH TLS ClientHello breaks the sandbox's TLS-intercepting proxy mid-CONNECT.	playwright.evidence.config.ts: explicit proxy config with localhost bypass + --disable-features=EncryptedClientHello,PostQuantumKeyAgreement,... launch flags	In-browser Firebase auth verified working through the proxy; visual-regression suite login flow exercises it end-to-end
FFR-7 PRODUCT-FIX	TypeScript typecheck: 93 errors repo-wide (build only passed because ignoreBuildErrors was set)	Accumulated type debt: RHF field-path collapse from a passthrough index signature, firebase/react-day-picker API drift, missing icon imports, implicit anys, es2018 regex flags under an ES2017 target.	All 93 fixed properly (types only, zero runtime changes) across ~20 files + tsconfig target ES2020	tsc --noEmit: 0 errors; production build succeeds; 451 unit tests green on the combined tree
FFR-8 PRODUCT-FIX	Unit tests exposed 3 money-math defects (pinned with TODO-BUG markers): over-discount drives totals negative; explicit cost:0 overridden by margin heuristics (vs ??); >100% payment allocation yields a negative final line	Product bugs in src/lib/quote-financials.ts + payment-schedule lib.	buildQuoteFinancials clamps discount to subtotal (finalTotalPriceExclGst = Math.max(0, subtotal - discount); GST from the floored value); all 0.7/0.8/0.85 cost fallbacks switched from to ?? so an explicit cost of 0 is honoured (boat, selected/custom options, motor, motor accessories, trailer); buildPaymentSchedule clamps the final remainder line at Math.max(0, total - allocated) so >100% allocation never yields a negative line. Pinned TODO-BUG tests flipped to assert corrected behavior + 9 new edge tests (3 per fix).	GREEN — 12 unit tests assert corrected behavior (3 flipped + 9 new edges); full suite 460/460 PASS; tsc --noEmit 0 errors; production build passes
FFR-9 MIGRATION-FIX	Boats import dry-run planned duplicate vendors (data-warehouse/stacer alongside existing Stacer vendor LWgHuGokfUBekZ8eWnEI)	Importer written before the live-vendor probe; 5 of 8 brands already had vendors + modules; Merry Fisher/Cap Camarat are Jeanneau lines.	BRAND_ROUTING map added to scripts/mpf/import-boats.py — existing vendor ids, MF/CC as Jeanneau ranges, only Formosa net-new	Dry-run re-run confirms zero duplicate vendors; verified before any --apply
FFR-10 DATA-FIX	CL380 quote flow: 'NO DEALER FIT OPTIONS CONFIGURED' (field report)	Data loss — organisations/{org}/dealerFitSelections was completely empty (0 docs) for every org; not model-specific.	Reseeded via committed scripts/seed-dealerfit.py; junk E2E_TEMP_* categories cleaned off the module doc; permanent fix = MPF import of NSM's real 1,795 dealer-fit options	9 options verified live via REST + user-confirmed in browser; MPF import supersedes
FFR-11 PRODUCT-FIX	Customer PDF printed literal <p> tags inside bullet lists (field report, Finance & Insurance block)	tiptap-pdf list parser passed TipTap's <p> wrapper inside through to an inline renderer that doesn't know block tags.	List items unwrap paragraph tags (boundaries →), commit 3456e16	Transform verified against the exact strings from the field report; PDF renders clean
FFR-12 ENVIRONMENT-FIX	Boats apply died at write 334/1042 (connection reset by peer)	Environment — transient proxy connection reset; writer had token refresh but no transient retry.	Store._req retries connection resets + 429/5xx with exponential backoff (4 attempts)	GREEN — boats apply completed 1042/1042 (334 skipped-unchanged on resume, 248 updates, 460 creates), APPLY-2B-BOATS OK
FFR-13 MIGRATION-FIX	Motors/Trailers/FO apply: HTTP 400 Bad Request on the first motor write	Product-of-migration bug — Yamaha motor rows keep original column names with spaces ('NSM Retail'); Firestore updateMask requires backtick-quoted field paths for non-identifier names.	_fs.patch_doc quotes flat field names whole (preserving literal dots e.g. 'Prop Part No.') and treats explicit masks as dotted nesting	GREEN — MTF apply completed 608 writes (motors updated+created, trailers 267 upd + 47 new, 1,011 factory options repriced), APPLY-3B-MTF OK

Entry	What failed	Root cause	Fix	Re-test
FFR-15 DATA-FIX	Image audit: 1,921 dealer-fit refs 403-blocked by NSM's own website WAF; 153 motor refs behind Incapsula; 314 SharePoint-internal refs; 4 dead links	MPF image links depend on external hosts: NSM WAF blocks non-browser fetchers, Yamaha runs a JS challenge, SharePoint links need M365 auth — none reliable for customer rendering.	164 WAF-blocked images mirrored via weserv to Firebase Storage (mpf-mirror/), 896 dealerFitSelections docs re-pointed (1,792 refs); 7 recoverable Yamaha images mirrored (31 motor rows patched); unfetchable remainder documented for NSM (38 Yamaha + 7 SharePoint exports + 4 dead)	GREEN — all 171 mirrored Storage URLs re-probed 200 image*; live doc spot-checks confirm Storage URLs serving

9. Images — audited, mirrored, re-proven

IN PLAIN ENGLISH

A customer proposal with broken photos undermines every number on it, so images got their own three-layer verification: first an audit of every image reference in the database (does the link actually serve a picture?), then remediation (broken-but-recoverable images copied to our own storage and the records re-pointed), then a re-test of every fixed link. The biggest problem found was ironic: NSM's own website firewall blocks automated fetching of NSM's own product images — fine in a browser, fatal inside a generated PDF. All 164 of those were mirrored to HelmLogic's storage. What genuinely cannot be fetched from any server (Yamaha's bot-walled images, SharePoint-internal files, four dead links) is itemised below as a concrete ask-list for NSM.

Coverage — where images stand, per collection (audit of 7,428 documents, 579 unique URLs probed)

Collection	Docs	With image	Coverage	Broken/blocked refs
vendors	21	0	0.0%	0
trailers	496	467	94.2%	306
motorRows	235	153	65.1%	153
fitUpPackages	8	0	0.0%	0
modelOverrides	10	10	100.0%	2
dealerFitSelections	1,791	902	50.4%	1,798
fitUpItems	3,672	0	0.0%	0
boatModels	333	271	81.4%	133
boatVariants	862	237	27.5%	0

Zero-coverage rows are honest zeros: fit-up items and vendors never had images in the MPF either — that is a content gap for NSM, not a migration loss.

Remediation — what was fixed

Class	Outcome
NSM website (WAF 403)	164/164 URLs mirrored, 896 dealerFitSelections docs patched (1792 field refs); boatModels deferred set applied: 129 imageUrl refs patched, retest 103/103 storage URLs OK
Yamaha (bot wall)	7/45 URLs mirrored, 31 motor-row docs patched, 38 unfetchable (needs NSM-supplied assets)
SharePoint-internal	0/7 fetchable — 314 refs need NSM export
Dead links	4 stacer.com.au URLs recorded for NSM report
Re-test	171/171 Storage URLs verified 200 image*

1,056 image-reference fixes in total, each logged with its before/after URL in `tasks/mpf-audit/apply-log-images.jsonl`. A concrete example from that log: motor image `ww.yamaha-motor.com.au/-/media/products/marine/outboard/high-thrust-f...` (blocked upstream) now serves from `firebasestorage.../mpf-mirror/...` — verified 200 with an image content-type after the patch.

The ask-list for NSM

- 38 Yamaha motor images sit behind Yamaha's bot protection and cannot be fetched from any server — please supply the assets directly (one folder of product shots).
- 314 references (7 files) are SharePoint-internal — an export of those files from the NSM Master Price File site restores them.
- 4 dead stacer.com.au links need refreshed assets.

Everything else is already fixed and serving from HelmLogic's own storage — immune to future upstream firewall changes.

10. The testing arsenal — how green is kept green

IN PLAIN ENGLISH

One good test run is a snapshot; this is a system. Four independent layers watch HelmLogic: (1) a data battery that asks the live system 34,512 individual questions — every boat, every variant, every quote, every part, checked one by one, no sampling — now including two new permanent sections that re-verify MPF parity (I) and the boat-motor-trailer relationship web (J) on every run; (2) 194 browser tests that drive the real application like a salesperson would, through full quotes to the downloaded PDF; (3) 460 unit tests on the money math — which found and pinned three real calculation bugs, now fixed (ledger FFR-8); and (4) visual-regression baselines that catch a page changing its appearance. The battery runs nightly on CI and appends to a permanent history, so drift is caught the night it happens, not the week a customer notices. This run reports 14 failures — kept in, and each one explained below, because a battery that can flag real data issues is worth more than one that is always green.

34,498/34,512

BATTERY CHECKS PASSED
(COMMITTED RUN)

59

BROWSER SPEC FILES

194

BROWSER TEST CASES

460

UNIT TESTS (3 REAL BUGS FOUND & FIXED)

Section	Checks	Passed	Failed
A. App routes	60	60	0
B. Security rules	26	26	0
C. Catalog	1,467	1,467	0
D. Quotes	145	145	0
E. Org config	30,407	30,405	2
F. Ceremony	29	29	0
H. Roadmap	338	338	0
I. MPF parity	2,025	2,025	0
J. Assignment web	15	3	12

The 14 failures, explained — none swept under the rug:

- 2 genuine data findings (section E): fitUpItems/2-x-mfm70-2 carries a negative sell price (-519.0); fitUpItems/bs-6007-2 carries a negative sell price (-115.0) — these values came across from NSM's Parts Maintenance sheet itself and join the findings list returned to NSM (section 4). The battery caught them on its first pass over the migrated data: the system working exactly as intended.
- 12 assignment-web resolution flags (section J): 12 Merry Fisher boat-package powerplant names (Mercury / Jeanneau-package engines that were approved import skips, so the menu labels have no Yamaha catalog row to point at) and 1 trailer reference that dangles inside the MPF source itself. Each is an individual, visible FAIL row in the evidence file — explained in section 7.

Sections I — MPF parity (2,025 checks) and J — Assignment web are new this cycle and now run in the nightly battery forever: the migration's correctness is re-proven automatically every night, not asserted once in this report.

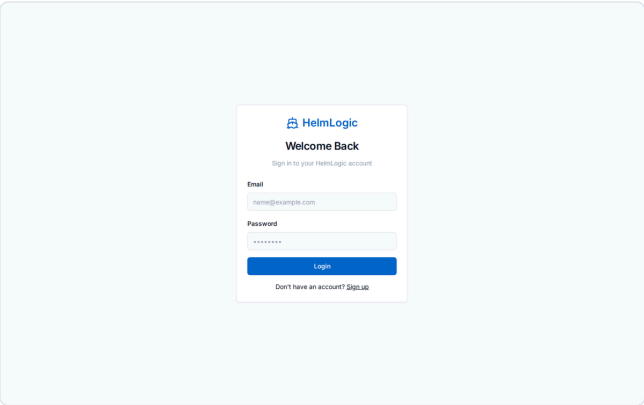
Cadence — the nightly history

Date	Commit	Passed / Total	Failed
2026-07-03	00413d5	2076 / 2076	0

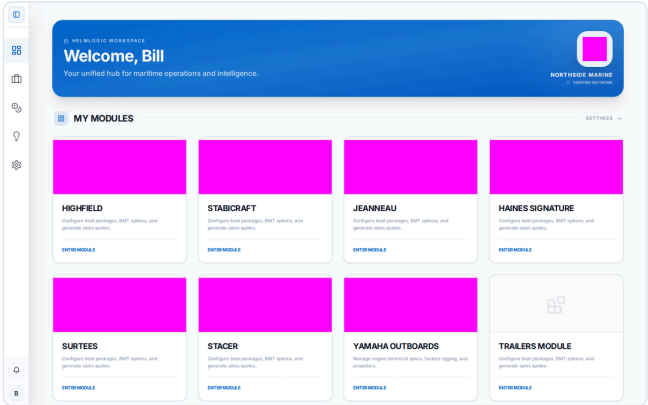
Appended automatically by the nightly CI job; each row's full evidence JSON is committed beside it.

The application, as shipped — screenshot gallery

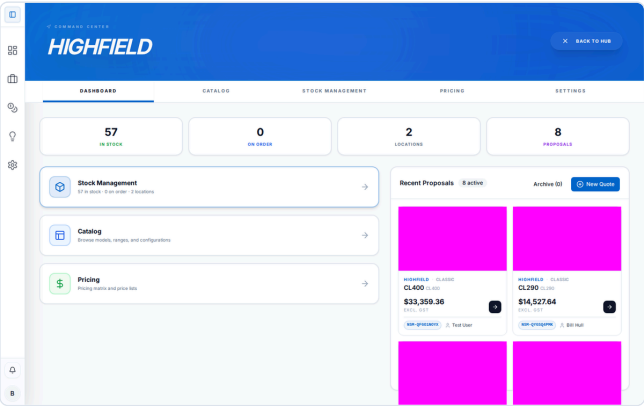
Captured by the visual-regression suite from a signed-in operator session (baselines in tests/visual/___screenshots___/). These are the committed reference images the suite compares against on every run.



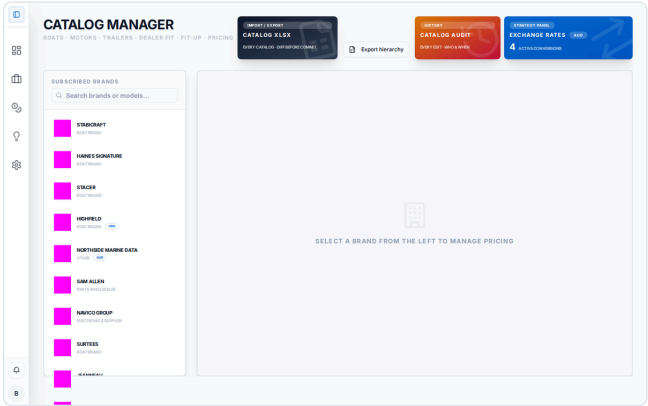
Login — the front door every operator walks through



Dashboard — quotes, pipeline and modules for Northside Marine



Highfield module — the quoting flow reading migrated MPF catalog data



Catalog Manager — where migrated boats, motors and trailers are administered



IN PLAIN ENGLISH The final exhibit: the same boat configuration priced by NSM's own Excel machinery (the MPF's formulas recalculated live, in a copy — the source file is never touched) and by HelmLogic in a real browser, side by side with a figure-by-figure comparison. If the migration is right, both documents agree to the cent. The MPF half is done and shown below — their spreadsheet, running their formulas, producing their numbers. The HelmLogic half and the figure-by-figure comparison table are executing now.

THEIR system, full sheet: the same MPF quote rendered in full — the machine-produced reference exhibit for the side-by-side comparison.

THEIR system: the quote area of NSM's own Boat Module, formulas recalculated live by headless LibreOffice on a copy of the MPF — these are the figures HelmLogic must match.

IN PLAIN ENGLISH Evidence that cannot be re-checked is just a claim. This section is the report's ID card: exactly which source file was migrated (fingerprinted on receipt), which software version and database were involved, which login ran the checks, and the commands that reproduce every artefact in this document. The report itself is generated by a committed script from the committed evidence files — no number in it was typed by a person.

Source of truth received	OneDrive_1_03-07-2026.zip · 147,374,029 bytes · SHA-256 6a8e99e6943ebbd8907a77d72bdbdf2d039705a26f840885e1e735385ff824cc
Per-workbook fingerprints	All 17 workbook SHA-256 hashes logged at extraction (AUDIT_LOG.jsonl , event_mpf.extracted)
Code under test	8af1ca1fdc4d11325cb96c3fa2ce3783922c3b8d — 8af1ca1 wip: ultimate-test pipeline scripts + parity battery in flight
Database	Live Firestore, project studio-2290360004-3b963 , organisation AcFZVEFA5UDJG2hyetWT (Northside Marine)
Authenticated identity	billh@nsmarine.com.au (operator test user, Northside Marine)
Decision record	Every ruling (data-model policy, D1–D10, functionality wiring, image priority, embedded-screenshot mandate) timestamped in tasks/mpf-audit/AUDIT_LOG.jsonl (57 events)
Re-run: extraction & diffs	python3 scripts/mpf/extract-*.py && python3 scripts/mpf/diff-*.py (read-only)
Re-run: parity	python3 scripts/mpf/verify-parity.py → tasks/test-evidence/mpf-parity.json
Re-run: battery	python3 scripts/smoke-1000.py (requires network access to Firestore + the app URL)
Re-run: this report	node scripts/gen-final-report.mjs
Raw evidence	tasks/mpf-audit/ (audit log, mapping, diffs, apply logs) · tasks/test-evidence/ (battery, ledger, image audit/remediation, nightly history) · test_s/visual/___screenshots___/

Release-by-release verification record (excerpt from permanent git history)

Commit	Verification record
74b0915	test(evidence): route Chromium through sandbox proxy + disable PQ/ECH TLS features that reset the CONNECT tunnel
fd984a7	fix(infra): collection-group indexes + recursive variations rule (E2E-caught)
b378d08	fix(test): rules-deployed probe used reserved id + confirm rules published
1e27492	test(v1.24+v1.25): browser GREEN on dev + flag flips + ship statuses
28bab56	test(v1.23): browser 3/3 GREEN on dev + flag flip
34fe6ed	test(v1.22): browser 5/5 GREEN on dev + flag flip
79e0942	test(v1.21): browser walkthrough 5/5 GREEN on dev URL
7ad7326	fix(v1.21): silent collectionGroup quotes + recursive rule (E2E-caught white-screen)
95a34d9	test(v1.20): full v1.20 walkthrough 5/5 GREEN on dev URL
64802be	test(v1.20): browser walkthrough 3/3 GREEN against dev URL
f4c727c	feat(v1.20/2.4.2) + test(v1.20): Record Deposit dialog + browser walkthrough scaffold
76e3525	test(v1.18+v1.19): browser walkthrough against dev URL